

Mutations Innate Immune System NOD/CARD15 Gene and Toll-Like-Receptor-4 Gene Are Associated with Increased Risk for Severe Acute Gvhd in Patients Who Underwent Allogeneic Transplantation.

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Abstract

The innate immune system detects invading pathogens through several pattern-recognition receptors and represents the first line of mucosal host defense. Mutations of the NOD2/CARD15 and Toll like receptor –2, –4, –9 genes have been associated with an increased incidence of inflammatory bowel disease due to a diminished response to bacterial cell wall products. Moreover, it was reported that mutations of NOD2 gene loci might be associated with an increased risk for TRM and severe GVHD in patients who underwent allogeneic blood stem cell transplantation. Here we analyzed 201 patients and their respective donors for NOD2, TLR-2, TLR-4 and TLR-9 mutations and correlated the results with the incidence of overall acute GVHD and intestinal GVHD. Further, we evaluated if the occurrence of NOD2/CARD15 and TLR allele mutations are accompanied with an increased risk for transplant-related mortality and overall survival.

Mutated alleles of the NOD2 gene were observed in 17% of the patients and 16% of the donors. NOD2 gene mutations in patients and donors together were associated with an increased incidence of severe acute GVHD grade III-IV (6/11 patients; 55% versus 35/145 patients; 24%). If a mutated allele of the NOD2 gene was found either in patients or in donors only no correlation with increased incidence of severe acute GVHD was seen in this study.

Patients with mutated alleles in TLR-4 genes (Asp299Gly and Thr399Ile) had a higher incidence of intestinal GVHD ($p < 0.02$), whereas mutations of donor alleles only had no influence on the occurrence of intestinal GVHD in this analysis.

No correlation with intestinal GVHD or overall acute GVHD was seen in patients with mutated alleles for TLR-2 and TLR-9. Overall survival, TRM and relapse risks was not influenced in none of the patients with mutated alleles of the innate immune system genes. Multivariate analysis including all potential factors, confirmed that mutations of NOD2/CARD15 in patients and donors together influenced the occurrence of severe acute GVHD, whereas the occurrence of intestinal GVHD was influenced in patients by mutated alleles of TLR-4 gene. But mutations in alleles of the NOD2 or TLR-2, -4, -9 had no influence on the outcome of allogeneic transplant (TRM and overall survival) in the multivariate analysis.

Since all patients in our institution had received an intestinal bacterial decontamination using metronidazole and ciprofloxacin after allogeneic stem cell transplantation, it might be speculative, if the reduction of concentrations of anaerobic bacteria in the intestine might have protected patients from the occurrence of increased TRM. Decontamination of anaerobic bacteria is associated with a reduced risk for severe acute GVHD as reported earlier (*Blood* 1999; 93:3267–75).

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